



Project 1

Domain: Tourism/Social Media

Techniques: NLP, Machine Learning

Title: Fake Review Prediction

Overview and Problem Statement:

One of the fields in which disinformation is seen often is hotel reviews, both for positive and for negative reviews. There may be an interest to spread positive or negative fake news about hotels to gain unfair competitive advantage or as (unethical) means for profitability. The main idea of this project is to reduce the impact of disinformation by building a machine learning model to predict fake reviews pertaining to the hotel industry. This will also involve using computational stylometry based analysis to detect the specific stylistic features that differentiate fake from real (truthful) reviews.

Dataset:

The Deceptive Opinion Spam corpus consists of truthful and deceptive hotel reviews of 20 Chicago hotels and contains 400 truthful positive reviews from TripAdvisor and 400 deceptive positive reviews from Mechanical Turk in addition to 400 truthful negative reviews from Expedia, Hotels.com, Orbitz, Priceline, TripAdvisor and Yelp and 400 deceptive negative reviews from Mechanical Turk. Each dataset consists of 20 reviews for each of the 20 most popular Chicago hotels. In total there are 1600 reviews, and the task is to classify the real vs fake hotel reviews using a classical machine learning or deep learning algorithm. Dataset source: <https://myleott.com/op-spam.html>

Specific Challenges:

1. Detection of author specific writing stylistic features to differentiate truthful vs fake reviews
2. Obtaining a good accuracy on new reviews from other websites

References:

1. Dataset link:
[https://cdn.iisc.talentsprint.com/CDS/Datasets/Deceptive Opinion Spam Corpus v1.4.zip](https://cdn.iisc.talentsprint.com/CDS/Datasets/Deceptive%20Opinion%20Spam%20Corpus%20v1.4.zip)
2. Pascucci et al., (2020): <https://aclanthology.org/2020.stoc-1.6.pdf>.
3. Kennedy et al., (2020). Fact or factitious? Contextualized opinion spam detection. *arXiv preprint arXiv:2010.15296*.



Project 2

Domain: Plant Pathology

Techniques: Deep Learning

Title: Image-based plant disease identification

Overview and Problem Statement:

Crop losses due to diseases are a major threat to food security every year, across countries. Conventionally, plant diseases were detected through a visual examination of the affected plants by plant pathology experts. This was often possible only after major damage had already occurred, so treatments were of limited or no use. Recently, access to smartphone based image capturing has highly increased amongst farmers and agriculturists. This has led to the successful adoption of plant disease diagnostic applications based on deep learning techniques. This is of immense value in the field of agriculture and an excellent tool for faster identification and treatment of crop diseases. It holds key importance in preventing crop based food and economic losses. The goal of this project is to build a convolutional neural network or to use transfer learning and develop a plant disease identification tool.

Specific Challenges:

1. Extending the applicability of the plant disease identification tool for farmers through the use of Indian regional languages [Optional challenge]

DataSet:

The PlantVillage dataset consists of 54303 healthy and unhealthy leaf images divided into 38 categories by species and disease. Dataset link:
<https://data.mendeley.com/datasets/tywbtsjrjv/1>

References:

1. Dataset link: <https://data.mendeley.com/datasets/tywbtsjrjv/1>
2. Dataset source details: https://www.tensorflow.org/datasets/catalog/plant_village
3. Liu and Wang (2021):
<https://plantmethods.biomedcentral.com/articles/10.1186/s13007-021-00722-9>
4. Mohanty et al., (2016):
<https://www.frontiersin.org/articles/10.3389/fpls.2016.01419/full>



Project 3

Domain: Food / Wellness industry

Techniques: CNN based Image segmentation

Title: Food Image Segmentation

Overview and Problem Statement:

Worldwide, obesity has nearly tripled since 1975. In 2016, more than 1.9 billion adults, 18 years and older, were overweight (WHO sources). In such a situation, documenting dietary caloric intake is crucial to manage weight loss. Food image segmentation is a critical and indispensable task for developing health-related applications such as automated estimation of food calories and nutrients as a means for dietary monitoring. One of the challenges in this area is the improvement of accuracy in dietary assessment by food image analysis. However, how to derive the food information (e.g., food type and portion size) from food images effectively is a challenging task and an open research problem. In this project, participants are expected to make a model that can segment the food components present in an input food image and build an application that can predict the food class and the food portions from it.

Specific Challenges:

1. The complex appearance of food makes it difficult to localize and recognize ingredients in food images, e.g. the components may overlap with one another in the same image, and the identical ingredient may appear distinctly in different food images.

Data Description:

Food recognition dataset (Ciocca et. al., 2017) is one of the few publicly available, pixel segmented datasets on food. It contains 1,027 images of food on trays, with 73 classes of food and 3,616 labelled instances of food. The tray images have been manually segmented using carefully drawn polygonal boundaries.

Application:

It has many applications such as calorie estimation, diet management, industrial compliance and customer satisfaction.

References:

1. Ciocca et al., 2017: Food Recognition: A New Dataset, Experiments, and Results
2. Yang, 2020: Food Image Segmentation with fast.ai
3. Liu et al., 2016: DeepFood: Deep Learning-based Food Image Recognition for Computer-aided Dietary Assessment



Project 4

Domain: Financial Services

Techniques: NLP, Conversational AI, ML, LLMs

Title: Personal AI Financial Advisor for Retail Clients

Overview and Problem Statement:

Retail clients often lack affordable and accessible personalized financial advice, leading to challenges in goal-based financial planning, insurance advisory, and portfolio management. This project focuses on developing a conversational AI system powered by LLMs to address these issues. The system will provide tailored financial recommendations, bridging the financial literacy gap and empowering users to make informed decisions.

Specific Challenges:

- Handling diverse user demographics and financial goals.
- Integrating real-time financial market data into conversational AI.
- Ensuring regulatory compliance and data security.

Data Description:

- Financial Planning Dataset:
 - Includes user demographics, financial goals, income, expenditure, and portfolio details.
 - Source: Simulated datasets or publicly available financial datasets (e.g., Kaggle, UCI Repository).
(<https://www.kaggle.com/datasets/bukolafatunde/personal-finance/data>)
(<https://www.kaggle.com/datasets/abhilashayagyaseni/personal-finance-dataset/data>)
- Conversational Data:
 - Pre-trained datasets like Microsoft Dialogue Corpus fine-tuned for financial use cases.
 - Source: (<https://github.com/emilyallaway/zero-shot-stance/tree/master/data>)
- Insurance and Financial Products Dataset:
 - Coverage options, premiums, real-time stock prices, and market trends.
(<https://www.kaggle.com/datasets/mrmori/insurance-recommendation/data>).



- Insurance Dataset: (<https://github.com/AntonUBC/Prudential-Life-Insurance-Assessment/tree/master/data/data>)

The Description of each variable in the dataset is given below.

Id- A unique identifier associated with an application; Product_Info_1-7- A set of normalized variables relating to the product applied for; Ins_Age- Normalized age of applicant; Ht- Normalized height of applicant; Wt- Normalized weight of applicant; BMI- Normalized BMI of applicant; Employment_Info_1-6- A set of normalized variables relating to the employment history of the applicant; InsuredInfo_1-6- A set of normalized variables providing information about the applicant; Insurance_History_1-9- A set of normalized variables relating to the insurance history of the applicant; Family_Hist_1-5- A set of normalized variables relating to the family history of the applicant; Medical_History_1-41- A set of normalized variables relating to the medical history of the applicant; Medical_Keyword_1-48- A set of dummy variables relating to the presence of/absence of a medical keyword being associated with the application; Response- This is the target variable, an ordinal variable relating to the final decision associated with an application.

- Alpha Vantage Financial Market Data:
Alpha Vantage is a provider of financial market data APIs that offer real-time and historical data for various financial instruments such as stock prices, technical indicators, sector performance, exchange rates, and more. This data should be utilized to train and fine-tune the Large Language Model (LLM). Real-time financial data and market news and trends should be incorporated to provide current and accurate financial advice.

Source: Alpha Vantage (<https://www.alphavantage.co/>)

Methodology:

- Extract user demographics, financial goals, income, and portfolio details; clean and normalize datasets.
- Fine-tune pre-trained LLMs using financial dialogue datasets to understand queries like "How can I save for retirement?" or "Should I invest in bonds?"
- Train ML models (e.g., Random Forest, XGBoost) to provide goal-based financial advice and insurance recommendations.
- Incorporate real-time market data from Alpha Vantage for contextual recommendations.
- Build a mobile-friendly conversational AI tool with personalized goal planning and portfolio analysis features.



Project Deliverables:

1. A conversational AI system capable of engaging retail clients with financial advisory discussions.
2. Goal-planning recommendations based on user profiles and inputs.
3. A portfolio analysis tool with actionable suggestions for realignment.
4. Integration of insurance advisory into the conversational system.

Applications:

This project democratizes financial advisory services, enabling personalized financial planning, portfolio optimization, and insurance recommendations at scale.

References:

- "Integrating Modern Conversational AI Architectures in FinTech: Advancements, Applications, and Challenges." (<https://www.ijert.org/research/integrating-modern-conversational-ai-architectures-in-fintech-advancements-applications-and-challenges-IJERTV13IS070017.pdf>)
- "AI-Driven Financial Analysis: Exploring ChatGPT's Capabilities and Challenges." (https://www.researchgate.net/publication/380873806_AI-Driven_Financial_Analysis_Exploring_ChartGPT's_Capabilities_and_Challenges)



Project 5

Domain: Financial Services

Techniques: Generative AI, Sentiment Analysis, Data Visualization, ML

Title: Database and GenAI-Powered Visualization Tool for Day Traders

Overview and Problem Statement:

Day traders face difficulties processing vast amounts of real-time data, including market sentiment, stock movements, and financial reports. This project aims to build an AI-powered dashboard to automate sentiment analysis, provide actionable buy/sell recommendations, and present interactive visualizations for quick, informed decision-making.

Specific Challenges:

- Real-time processing and integration of large datasets.
- Ensuring high accuracy in financial sentiment analysis.
- Developing low-latency interactive visualizations.

Data Description:

- **Market Sentiment Data:** News headlines, social media posts, and sentiment scores.
 - Source: Public datasets such as [Twitter and Reddit Sentimental analysis Dataset](#) and [Financial Sentiment Analysis](#)
- **Stock Market Data:** Historical prices, trading volumes, and sector performance.
 - Source: [Yahoo Finance Dataset \(2018-2023\)](#), [Yahoo Finance Historical Data](#) or Quandl's Stock Price Data [Quandl WIKI Prices US Equites](#)
- **Company Financials:** Annual reports and key financial metrics.
 - Source: [SEC Financial Statement Data Sets](#) or [Kaggle's Financial Data Repositories](#).

Methodology:

- Merge market sentiment, stock prices, and financial data into a unified dataset.
- Fine-tune financial NLP models like FinBERT to classify market sentiment as bullish, bearish, or neutral.
- Use ML models (e.g., LightGBM) to analyze trading volume trends, sentiment scores, and stock indicators.
- Create a real-time dashboard with GenAI-enabled natural language queries, interactive charts, and sector comparisons.



Project Deliverables:

1. A fully functional AI-driven dashboard for day traders.
2. Sentiment analysis feature to process news and social media in real time.
3. Interactive buy/sell recommendations with risk scores.
4. Summaries of company financial statements.

Applications:

The tool supports day traders by reducing decision-making time through actionable insights, sentiment analysis, and financial summaries.

References:

- Yang, Liu, and Wang (2023): "FinGPT: Open-Source Financial Large Language Models." (<https://arxiv.org/pdf/2306.06031>)
- Financial PhraseBank: Sentiment Analysis of Financial News Using BERT Models. (https://www.researchgate.net/publication/382309719_SENTIMENT_ANALYSIS_OF_FINANCIAL_NEWS_USING_THE_BERT_MODEL)



Project 6

Domain: Education

Techniques: Siamese Text Similarity Model, NLP, Deep Learning

Title: Automated Answer Validation for Science Question Answering using Siamese Text Similarity Model

Overview and Problem Statement:

In the domain of scientific question answering, validating answers and providing accurate feedback is critical for effective learning. The goal of this capstone project is to develop an automated answer validation system using a Siamese text similarity model. The system will compare student responses with the correct answer and distractors to determine the level of correctness and provide appropriate feedback. The automated answer validation system for science question answering will benefit educators and students in science-related subjects. It will streamline the assessment process, reduce manual effort, and ensure consistent evaluations, leading to improved learning outcomes in the science domain.

Specific Challenges:

1. Implementing a robust Siamese text similarity model for answer validation in the science domain.
2. Preprocessing and structuring the dataset from the provided SCIQ dataset to create meaningful pairs of student responses and correct answers.
3. Handling varying lengths of text data during model training and inference.
4. Addressing potential semantic ambiguities and domain-specific challenges in science questions and responses.

Data Description:

The SciQ dataset contains 13,679 crowdsourced science exam questions about Physics, Chemistry and Biology, among others. The questions are in multiple-choice format with 4 answer options each. For the majority of the questions, an additional paragraph with supporting evidence for the correct answer is provided. (string).

- answer1: One of the candidate answers for the question (string).
- answer2: One of the candidate answers for the question (string).
- answer3: One of the candidate answers for the question (string).
- answer4: One of the candidate answers for the question (string).
- correct_answer: The correct answer for the question (string).
- support: The supporting text for the question (string).



Methodology:

- Combine the question and supporting text to create meaningful context for answer validation in the science domain. Structure the data to form pairs of student responses and correct answers, along with corresponding labels (0 for incorrect and 1 for correct).
- Implement a Siamese neural network architecture using NLP libraries and deep learning frameworks. Train the model on the pairs of student responses and correct answers to learn the underlying text representations
- Develop an interface for educators to input student responses and query the Siamese text similarity model for validation in the science domain.
- Fine-tune the model and update the answer validation system based on the evaluation results.

References:

1. SCIQ (Science Question Answering) dataset from Kaggle:
<https://www.kaggle.com/datasets/thedevastator/sciq-a-dataset-for-science-question-answering>
2. Siamese network for image and text similarity:
<https://medium.com/@prabhnoor0212/siamese-network-keras-31a3a8f37d04>



Project 7

Domain: Interior Design

Techniques: Stable Diffusion, Transformers

Title: AI-Driven Interior Design Using Stable Diffusion and Transformers

Problem Statement:

AI-driven innovation in creative industries such as design offers a very good potential. Be it architectural design, landscape design or interior design, the possibilities are endless. Such generated designs have the potential to drive rapid growth and profits in the interior design industry. The goal of this project is to generate realistic and novel interior room designs by leveraging Stable Diffusion and Transformer models, trained on the IKEA Interior Design Dataset. These generated designs can revolutionize the interior design industry by providing rapid, high-quality design options.

Specific Challenges:

1. Distinguishing a good design from a poor one or developing an evaluation metric is a complicated problem because good design is 'subjective'.
2. Ensuring the stable and reliable training of Stable Diffusion models to avoid issues like generating nonsensical designs.
3. Integrating Transformer models to effectively understand and apply design principles and context.

Dataset Description:

The dataset was collected from IKEA.com website for the purpose of building Style Search Engine (note: only for non-commercial use). It consists of: 2193 object (product) photos, 298 context (room scene) photos in which those objects appear, text descriptions for products, ground truth information on which items appear in which rooms.

Applications:

1. Web/app based designing tool for designers and real estate market - similar to Heavenly, Planner 5D
2. Automated interior design suggestions for enhancing user experience and creativity.

References:

1. Image to Image using Artificial Intelligence — Automating Room Interior Design, David Oluyale: <https://medium.com/@oluyaled/image-to-image-using-artificial-intelligence-automating-room-interior-design-5351bdfdd4bf>



2. Integrating aesthetics and efficiency: AI-driven diffusion models for visually pleasing interior design generation:

[https://www.researchgate.net/publication/378154158 Integrating aesthetics and efficiency AI-driven diffusion models for visually pleasing interior design generation](https://www.researchgate.net/publication/378154158_Integrating_aesthetics_and_efficiency_AI-driven_diffusion_models_for_visually_pleasing_interior_design_generation)

3. Tautkute et al., 2017, ACICS, IKEA: Interior Design Dataset

4. Denoising Diffusion Probabilistic Models, Ho et al., 2020:

<https://arxiv.org/abs/2006.11239>

5. Attention is All You Need, Vaswani et al., 2017: <https://arxiv.org/abs/1706.03762>



Project 8

Domain: Market analytics and segmentation

Techniques: Data Engineering, ML, Google Data Studio

Title: Big data analytics and clustering of Farmers markets

Overview and Problem Statement:

Farmer's markets have emerged as an alternative and an improvement to grocery stores in the US. The Farmers' markets are a great place where people can consume nutritious fresh food while helping small family farms. Here, using a huge Farmer Market dataset, business insights in products and sales will be obtained through Big Data Analytics and market segmentation.

Specific Challenges:

1. Find important market insights through Big Data Analytics.
Eg. Use Hadoop MapReduce to find the US state which has the lowest number of Farmer markets. Find the top 5 states that have the maximum number of products available. Assign ratings 1-5 for all the states based on the availability of products
2. Provide analytics to investors on states where new Farmer markets could be setup and prove to be profitable with a specific set of grocery products. Integrate other geographic/ state wise datasets eg. population/weather/average income if required to make better predictions
3. Clustering/market segmentation of the Farmer Markets to understand their distribution in terms of products, regions and sale

Data description:

The Farmers Market dataset is from the US data.gov site. This USDA National Farmers Market Directory data contains information on all registered farmers' markets in the US. Each row contains the information on the farmers market's name along with its city, county, types of food products sold, and the accepted forms of payment.

Application:

1. Obtaining business intelligence for investments in the Food retail industry
2. Obtaining competitive intelligence of understanding a company's industry and industry rivals so the company can make better business decisions

References:

1. USDA Farmers Markets: <https://www.ams.usda.gov/local-food-directories/farmersmarkets>
2. Dataset source: <https://data.world/johnsnowlabs/us-farmers-market-location>
3. Data Visualization: <https://www.susielu.com/data-viz/farmers-markets>



Project 9

Domain: Supply chain – Food Delivery

Techniques: ML Algorithms

Title: Driver Demand Prediction for Optimal Food Delivery Charges

Overview and Problem Statement

The food delivery industry relies heavily on the efficiency of its delivery processes, where timely and accurate deliveries significantly impact customer satisfaction and overall experience. This project focuses on developing an effective predictive model that accurately estimates the time it takes for orders to reach customers. Delivery time is influenced by various factors, including the delivery person's age, ratings, geographic coordinates, and time-related variables. The successful implementation of this food delivery time prediction model will enable businesses to optimize their operational processes, leading to more accurate delivery time estimates and an improved overall delivery experience for customers. The aim is also to predict the demand for delivery drivers in specific regions and times, by analyzing order requests, driver activity, and related parameters, thereby optimizing delivery charges, ensuring consistency and minimizing customer drop-offs.

Dataset:

The Kaggle [DeliverOnTime: Food Delivery Duration Predictor](#) consists of the features 'ID', 'Delivery_person_ID', 'Delivery_person_Age', 'Delivery_person_Ratings', 'Restaurant_latitude', 'Restaurant_longitude', 'Delivery_location_latitude', 'Delivery_location_longitude', 'Order_Date', 'Time_Orderd', 'Time_Order_picked', 'Weatherconditions', 'Road_traffic_density', 'Vehicle_condition', 'Type_of_order', 'Type_of_vehicle', 'multiple_deliveries', 'Festival', 'City', 'Time_taken(min)'

Challenges:

- Handling potential data inconsistencies, such as missing data or outliers.
- Predicting sudden surges in order demand, which may not always follow historical patterns.
- Geographical feature engineering
- Optional: Real-time driver demand prediction without compromising the accuracy or speed of the system.

Additional Reading:

[Urban distribution of short-term food delivery demand](#)



Project 10

Domain: Human Resources and Technology

Techniques: Large Language Model (LLM) Fine-Tuning, Natural Language Processing (NLP), Text-to-Speech, Video Generation

Title: Automated Video Creation from Resumes Using GPT

Overview and Problem Statement: Creating a concise and engaging video summary of a resume can significantly enhance the recruitment process. Traditional methods of resume screening are time-consuming and may overlook key aspects of a candidate's profile. This project addresses the need for an automated solution to convert text-based resumes into 1-minute video summaries, effectively showcasing the candidate's skills, experiences, and achievements. The project involves fine-tuning a GPT model to generate a script from a resume and utilizing text-to-speech and video generation technologies to create the video.

Dataset: The primary dataset used for this project will be the [Resume Dataset](#) from Kaggle. This dataset contains a diverse collection of resumes, covering various industries and roles. The resumes are annotated to highlight key sections such as contact information, summary, skills, work experience, education, and achievements. This comprehensive dataset will be instrumental in training the model to accurately interpret and summarize resume content.

Specific Challenges:

- Ensuring the model accurately interprets the context and importance of various resume sections to create a coherent and relevant script.
- Tailoring the video script to reflect the unique strengths and experiences of each candidate.
- Generating human-like and engaging scripts that effectively summarize the resume content.
- Integrating text-to-speech technology to produce natural-sounding narration and combining it with relevant visual elements to create a professional video.

References:

1. Text-to-Video Generative AI Models: The Definitive List: <https://aibusiness.com/nlp/ai-video-generation-the-supreme-list>
2. Everything to Know About OpenAI's New Text-to-Video Generator, Sora: <https://www.scientificamerican.com/article/sora-openai-text-video-generator/>
3. Resume Video Maker: <https://www.steve.ai/resume-video-maker>



Project 11

Domain: E-commerce and Fashion

Techniques: Deep Learning (Stable Diffusion, Transformer)

Title: Fashion Compatibility Prediction

Overview and Problem Statement:

The fashion domain is a very important and lucrative application of computer vision. According to a recent study by Statista, the fashion industry's worth was estimated to be \$1.5 trillion in 2020 and it keeps growing, representing a huge market for garment companies, designers, and e-commerce entities. Fashion image retrieval and fashion image attribute learning have been the two main areas of study in this domain. The goal of this project is to compose or predict fashion outfits automatically, working to address challenges in compatibility and aesthetics using advanced deep learning models like Stable Diffusion and Transformers.

Specific Challenges:

1. Learn compatibility relationships among fashion items to facilitate effective fashion recommendation using a Transformer model. Transformers are highly effective in capturing contextual relationships in sequential data.
2. Utilize Stable Diffusion for generating high-quality, realistic images of fashion outfits that follow compatibility criteria.
3. Note that this project is compute intensive. This might require the use of AWS Sagemaker free/purchased account or equivalent high-performance computing resources.

Data Description:

Polyvore is a popular fashion website, where users create and upload outfit data. These fashion outfits contain rich multimodal information like images and descriptions of fashion items, number of likes of the outfit, hash tags of the outfit, etc. Researchers have utilized this information for various fashion tasks. Therefore, here, a curated part of the Polyvore dataset (made available for research purpose through the ACM MM'17 paper "Learning Fashion Compatibility with Bidirectional LSTMs" [[paper](#)] [[code](#)]) will be used. It contains 164,379 items (each item contains a pair - product image and a corresponding text description). The average number of fashion items in an outfit is 6.5. The fashion-compatibility-prediction.txt contains ~7,000 outfits, where 4,000 are incompatible and 3,000 are compatible. In each line the first number indicates the compatibility (1 is compatible, 0 is not) followed by a sequence of fashion items consisting the outfit.

Applications:

1. Outfit match recommendation



2. Accessories recommendation
3. Realistic generation of fashion outfits
4. Personalized fashion recommendations

References:

1. Polyvore dataset: <https://github.com/xthan/polyvore-dataset>
2. High-Resolution Image Synthesis with Latent Diffusion Models, Rombach et al., 2021: <https://arxiv.org/abs/2112.10752>
3. Virtual Fashion Designer: <https://www.e2enetworks.com/blog/fine-tuning-stable-diffusion-to-create-a-virtual-fashion-designer-for-customers>



Project 12

Domain: Automotives & Insurance

Techniques: Deep Learning (Transfer Learning, Mask R-CNN)

Title: Automated car exterior damage assessment

Overview and Problem Statement:

Automated detection of car exterior damages and subsequent estimation of damage severity is of immense value for car dealers and insurance providers as it eliminates the manual process of damage assessment. This project is focussed on the detection of car damage and estimating its severity by using computer vision based Mask R-CNN technique. Mask R-CNN is an instance segmentation model that allows us to segment individual objects within a scene, regardless of whether they are of the same type.

Specific Challenges:

1. Data collection and annotation: Collect more car damage images under different weather conditions and different levels of illumination, enhance the data, improve the edge-contour enhancement of images, and make the masking of the damaged areas of the car more accurate.
2. Precise object detection
3. Robustness
4. Compute intensive. This might require the use of AWS Sagemaker free/purchased account.

Data Description:

1. Use web scraping from google images for downloading damaged and undamaged car images. Also include the COCO car damage dataset.
2. Annotate the images based on damage severity

Application:

1. Automated car damage assessment tool
2. Automated insurance claim prediction tool

References:

1. Sourish Dey, TowardsDataScience: Detecting car exterior damage
2. Priya Dwivedi, AnalyticsVidhya: Mask R-CNN model for detecting car damage
3. Zhang et al., 2020: <https://ieeexplore.ieee.org/document/8950115>



Project 13

Domain: R & D

Techniques: Large Language Model (LLM) Fine Tuning, Computer Vision, NLP

Title: Gemini-SCICAP: Enhancing Scientific Figure Captioning with a Language Model

Overview and Problem Statement:

Scientific figures often contain crucial information, and providing accurate captions is essential for better comprehension. Existing generic captioning models may not capture the specialized terminology and context found in scientific literature. This project addresses the need for a dedicated model for scientific image captioning. This project involves fine tuning the Gemini Large Language Model (LLM) to generate accurate and contextually relevant captions for scientific figures. A model capable of understanding and describing complex scientific visuals will be created, combining the power of NLP with computer vision

Dataset

SCICAP (Scientific Captioning) is a large-scale image captioning dataset containing real-world scientific figures and captions. The dataset is constructed using over two million images from more than 290,000 papers collected and released by arXiv. It covers a wide range of scientific domains, making it a comprehensive resource for training and evaluating the model.

Specific Challenges

- **Scientific Terminology:** Adapting the language model to understand and generate captions with domain-specific scientific terminology.
- **Complex Visuals:** Handling intricate scientific visuals that may include charts, graphs, and diagrams.
- **Contextual Understanding:** Ensuring the model captures the context of the scientific content to provide informative and coherent captions.
- **Multi-Modal Learning:** Integrating both text and visual information for effective image captioning.

References

1. Gemini: A Family of Highly Capable Multimodal Models
2. Gemini: An Overview of Multimodal Use Cases
3. SCICAP Dataset: A Large-Scale Scientific Image Captioning Benchmark



Project 14

Domain: Music Industry

Techniques: Association Rule Mining, Collaborative Filtering

Title: Market Basket Analysis for Personalized Music Recommendations

Overview and Problem Statement:

Music platforms with a large user bases often invest more in advanced recommender systems to cater to diverse user preferences. Competition among music streaming services drives innovation in recommender systems. Companies strive to differentiate themselves by offering more accurate and personalized recommendations, which can impact the overall market value. This project aims to create personalized music recommendations by employing Market Basket Analysis on the LFM-2b dataset. Using Data Mining, Market Basket Analysis and Recommender System Techniques, participants will utilize methodologies to uncover user behaviour patterns, fostering a deeper understanding of individual music preferences.

Dataset:

The LFM-2b dataset (<http://www.cp.iku.at/datasets/LFM-2b/>) contains the listening records of over 120,000 users of the music platform Last.fm. These users provide a total of more than two billion individual listening events that span a time range of over 15 years, from February 2005 until March 2020. These listening events refer to a total of 50 million distinct tracks of 5 million distinct artists. Beside the common metadata (i. e., artist and track name), LFM-2b contains additional information both regarding the users and items. This includes the demographic information of users, namely country, gender, and age, and the fine-grained genre and style of items together with the vector embeddings of their lyrics.

Specific Challenges

- Identifying frequent itemsets: Discovering commonly co-occurring music tracks in user listening histories and generating Association Rules to extract meaningful associations to understand music preferences.
- Utilizing user behavior patterns for personalized music recommendations through Collaborative filtering
- Optional: Apply matrix factorization techniques like Singular Value Decomposition (SVD) for latent factor modeling.

Reference:

LFM-2b Dataset Publication: <https://humrec.github.io/publication/schedl-chiir-2022/schedl-chiir-2022.pdf>



Project 15

Domain: Real Estate

Techniques: EDA, ML, DL, Time Series Analytics

Title: Market House price trends and prediction for metropolitan cities of India

Overview and Problem Statement:

Accurately predicting house prices is a challenging task. Potential home buyers assess multiple factors such as the size of the property, the location, road density, vicinity to offices, schools, stores, parks, restaurants, hospitals, airport, public transport etc. The goal of this project is to perform exploratory data analysis, time series analysis and predictive modelling for house prices in the metropolitan cities of India: Mumbai, Delhi, Bengaluru, Kolkata, Hyderabad and Chennai. This will involve deriving house price analytical insights through EDA and improving the accuracy over the current benchmarks using feature engineering (including geocoding) and model tuning.

Specific Challenges:

1. Feature engineering and model tuning
2. High predictive accuracy

Data Description:

1. For capturing house pricing trends over the years, National Housing Bank (NHB) data comprising of city wise housing price indices is available. It tracks changes in housing prices, measured against a base period and provides quarterly data on housing prices up to a neighbourhood level.
2. For predictive modelling, the dataset to be used is 'The housing prices in Metropolitan areas of India' (Kaggle). This dataset comprises of web-scraped data (~4000-7000 records per city) including the collection of prices of new and resale houses located in the metropolitan areas of India and the amenities provided for each house.

Applications:

Web based tool or mobile application for house price estimation, comparison and urban residential amenities assessment in metropolitan cities. Useful for individual home buyers and real estate businesses.

References:

1. Dataset: National Housing Bank - Residex
2. Dataset: Housing prices in Metropolitan areas of India
3. Sarkar and Purohit, Reuters, 2022. India house price rises pick up pace
4. Thakur and Satish, IRJET, 2021. Bangalore House price prediction



Project 16

Domain: Sports

Techniques: Statistics, Probability, Machine Learning, Data Visualization, Big Data

Title: Exploratory and predictive data analytics on the Indian Premier League (IPL) dataset with LLM Integration

Overview and Problem Statement:

The Indian Premier League (IPL) is a professional Twenty20 cricket league, contested by eight teams based out of eight different Indian cities. The league was founded by the Board of Control for Cricket in India (BCCI) in 2007. IPL is one of the most lucrative and powerful cricket tournaments for BCCI as well as the IPL players. The huge data collected from IPL matches over 10 years presents a vast opportunity for data analytics and insights from predictive studies. The goal of this project is to obtain statistical and predictive insights from the IPL dataset. This project will involve the use of statistical, probabilistic, machine learning, big data and data visualization tools to derive valuable information and predictive insights by integrating an LLM interface to allow users to interact with the data. A data dashboard of the analytics should also be created. The project should focus on integrating a Large Language Model (LLM) interface with the data analytics dashboard. This interface will allow users to ask natural language questions and receive statistical insights, making the data analysis more accessible and intuitive. This LLM-driven approach will not only facilitate traditional data analysis but also enhance user interaction, enabling more profound and dynamic insights into the IPL data.

Data Description:

The IPL Data till 2017 dataset includes detailed ball-by-ball data of all IPL matches up to the 2017 season. It consists of five files with 97 columns (attributes) in total:

- Ball_by_Ball.csv
- Match.csv
- Player_match.csv
- Player.csv
- Team.csv

By using the IPL dataset and incorporating LLM for a conversational interface, this project aims to create a powerful, user-friendly tool. The project ensures that the solution remains robust, data-driven, and user-centric, aligning with the evolving needs of the sports analytics field.

Applications:

1. Providing a statistical and data-driven basis for team formulation.



2. Developing in-game strategies.
3. Making real-time outcome predictions.
4. Gaining insights into cross-team player strengths and weaknesses.

References:

1. Data.world: IPL Data till 2017
2. Srinivasan, 2018: IPL and Big Data Analytics: A match made in heaven?
3. Kargal, 2021: IPL 2021: How data analytics is changing cricket



Project 17

Domain: Customer Service and Conversational AI

Techniques: Natural Language Processing (NLP), Large Language Models (LLMs), Sentiment Analysis, Intent Recognition, Topic Modeling

Title: Customer Conversational Intelligence Platform Powered by an LLM Agent

Overview and Problem Statement:

This project aims to develop a state-of-the-art Customer Conversational Intelligence Platform powered by a Large Language Model (LLM) agent. The LLM's advanced language understanding will drive the analysis of customer interactions across diverse channels (chatbots, call centers, email, social media). The platform will extract actionable insights from this data, enabling businesses to optimize customer service processes and significantly enhance the overall customer experience.

Datasets:

Name: Relational Strategies in Customer Interactions (RSiCS)

- Description: This dataset contains a corpus for improving the quality and relational abilities of Intelligent Virtual Agents.
- Link: [Link to the dataset](#)

Name: 3K Conversations Dataset for ChatBot from Kaggle

- Description: The dataset includes various types of conversations such as casual or formal discussions, interviews, customer service interactions, and social media conversations.
- Link: [Link to the dataset](#)

Name: Customer Support on Twitter Dataset from Kaggle

- Description: This is a large corpus of tweets and replies that can aid in natural language understanding and conversational models.
- Link: [Link to the dataset](#)

Challenges:

1. Data Collection: Gather customer conversations from diverse sources like voice calls, chat transcripts, emails, and social media interactions.
2. Use LLM-Agent for:



- a. Sentiment Analysis – accurate detection of customer emotions (positive, negative, neutral) and granular sentiment categories (frustration, satisfaction, inquiry, etc.) throughout conversations.
- b. Intent Recognition - understanding the underlying purpose behind customers' queries, enabling tailored responses and resolutions
- c. Topic Modeling - discovering recurring themes and patterns within conversations, highlighting trending issues, feedback topics, and potential areas for improvement.
- d. Agent Performance Evaluation - Analyzing agent interactions to provide constructive feedback, identifying training needs, and recognizing exceptional service.
- e. LLM-Driven Real-time Recommendations - Empowering agents with suggestions for next-best actions or responses during active conversations, optimizing outcomes.

Methodology:

Select GPT2/GPT3/GPT3.5, fine-tune the LLM agent extensively on a large dataset of customer conversations annotated for sentiment, intent, topics, etc. Develop ML algorithms to support the LLM agent. The primary focus will be on the LLM's ability to perform sentiment analysis, intent recognition, topic modeling, and agent performance assessment. Utilize platforms like SageMaker or equivalent to automate the ML workflow.

Example:

Customer: Hello, I ordered a laptop from your website, and it's been a week, but I haven't received it yet. Can you help me track my order?

Platform Analysis:

Categorization: Inquiry about order tracking.

Sentiment Analysis: Neutral sentiment.

Resolution Status: Unresolved.

Support Agent: Hi there! I apologize for the delay in your order. Could you please provide me with your order number? I'll check the status for you.

Customer: My order number is 123456789.

Platform Analysis:

Categorization: Providing order information.



Sentiment Analysis: Neutral sentiment.

Resolution Status: In progress.

Support Agent: Thank you for providing the order number. Let me check that for you.
[Platform sends a real-time request to the order tracking system]

Platform Analysis:

Real-time Analysis: The platform receives updated order tracking information. The laptop is currently in transit and is expected to arrive in two days.

Support Agent: Good news! Your laptop is on its way and should be delivered within the next two days. Here's your tracking number: ABC123XYZ. You can use this number to monitor its progress.

Customer: Thank you for the information. I appreciate your help.

Platform Analysis:

Sentiment Analysis: Positive sentiment.

Resolution Status: Resolved.

Support Agent: You're welcome! If you have any more questions or need further assistance, feel free to ask. Have a great day!

Significance:

The envisioned platform, with its real-time analysis capabilities, has the potential to revolutionize customer service, ultimately translating into greater customer satisfaction, increased operational efficiency, and a strengthened market position for businesses.

References:

1. Conversational Health Agents: A Personalized LLM-Powered Agent Framework, Mahyar Abbasian, Iman Azimi, Amir M. Rahmani, Ramesh Jain: <https://arxiv.org/html/2310.02374v4>
2. Building a Conversational AI Agent with Long-Term Memory Using LangChain and Milvus, Zilliz: https://medium.com/@zilliz_learn/building-a-conversational-ai-agent-with-long-term-memory-using-langchain-and-milvus-0c4120ad7426



Project 18

Domain: Healthcare

Techniques: Natural Language Processing, Generative AI

Title: Fine-Tuning the IndicTrans NMT Model for Healthcare Conversations

Overview and Problem Statement:

This project aims to enhance the performance of the IndicTrans Neural Machine Translation (NMT) model specifically for healthcare conversations between patients and hospital staff (doctors/nurses). The current model has very poor accuracy on medical terminology translations from Indian languages to English. The objective is to fine-tune the model using a dataset of healthcare conversations to improve its accuracy and fluency in translating medical terminology and patient queries from Indian languages to English. By fine-tuning the model on domain-specific data, we aim to optimize its performance for real-world healthcare communication scenarios.

Dataset:

Custom data is required for this project, wherein the patient's statement in a non-English Indian language (eg. Telugu) and its corresponding English translation are generated. These conversations may cover various medical scenarios, including patient inquiries, symptom descriptions, treatment discussions, and medical advice exchanges. [See a related (paid) dataset comprised of a healthcare discussion in Telugu here: <https://www.futurebeeai.com/dataset/speech-dataset/healthcare-call-center-conversation-telugu-india#contact-form>]. Other possibilities are to create synthetic medical conversation dataset (Telugu (or another Indian language) and English translation) using an LLM. For an example, see Fig. 1.

Methodology:

The IndicTrans NMT model, a pre-trained machine translation model tailored for Indian languages, will be selected as the base model for fine-tuning.

Dataset Preparation:

The healthcare conversation dataset will be preprocessed and formatted to prepare it for fine-tuning. This involves cleaning the data, tokenizing the text, and organizing it into a suitable format for training.



Fine-Tuning: The pre-trained IndicTrans NMT model will be fine-tuned on the healthcare conversation dataset.

Evaluation:

The fine-tuned model will be evaluated on a separate validation set to assess its performance in translating healthcare conversations accurately and fluently. Evaluation metrics such as BLEU score, accuracy, and fluency will be used to measure the model's effectiveness.

Model Deployment:

Once the fine-tuning process is complete and the model demonstrates satisfactory performance on the validation set, it will be deployed for use as a Whatsapp based app in healthcare communication scenarios.

By fine-tuning the IndicTrans NMT model on healthcare conversation data, this project aims to address the specific challenges and nuances of medical communication, thereby improving the accessibility and quality of healthcare services for Indian language speakers



Fig. 1 Example of synthetic data creation using ChatGPT (Telugu->English medical conversation samples)



Project 19

Domain: Legal Studies

Techniques: Natural Language Processing, Generative AI

Title: AI Patent Advisor: Leveraging Large Language Models for Patent Analysis and Technology Transfer Facilitation

Overview:

With the exponential growth of patent databases, extracting valuable insights and facilitating technology transfer has become increasingly challenging. This project aims to develop an AI-powered advisor that can analyze patents, provide comprehensive summaries, and recommend potential commercial applications and licensing opportunities. By fine-tuning the model on the AI-Growth-Lab patents and claims dataset, sourced from various patent repositories, the AI Patent Advisor will empower inventors, businesses, and legal professionals to navigate the complex landscape of patent law and innovation.

Dataset:

The AI-Growth-Lab patents and claims dataset (https://huggingface.co/datasets/AI-Growth-Lab/patents_claims_1.5m_train_test) will serve as the foundation for training the AI Patent Advisor. This dataset comprises a vast collection of patents and their associated claims, covering diverse technical fields and industries. With over 1.5 million patent documents, the dataset provides a rich source of information for training and fine-tuning the LLM model. Each patent document includes detailed descriptions, claims, and metadata, enabling comprehensive analysis and understanding of patented technologies.

Methodology:

- Data Collection and Preprocessing: Gather AI-Growth-Lab patents and claims, preprocess to clean noise, standardize formatting, and tokenize text
- Fine-tuning LLM: Utilize state-of-the-art LLM architecture, fine-tune on AI-Growth-Lab dataset for patent-specific language adaptation.
- Analyze patents, extract key concepts, and generate concise summaries for efficient knowledge extraction.
- (Optional) Implement semantic matching algorithms to map patented inventions to potential commercial applications and licensing opportunities.
- Design a user-friendly interface enabling users to input patents, explore summaries, and receive technology transfer recommendations.



- (Optional) Testing and Validation: Assess AI Patent Advisor's performance, including accuracy of summaries, relevance of transfer recommendations, and overall usability.
- (Optional) Deployment and Maintenance: Deploy the AI Patent Advisor, ensuring scalability and reliability, with protocols for regular maintenance and updates to align with evolving patent language and industry trends.

Challenges:

- Handling the complexity and variability of patent language and terminology; Ensuring the accuracy and relevance of patent summaries and technology transfer recommendations

Significance:

The AI Patent Advisor project holds immense potential to transform the landscape of patent analysis and technology transfer facilitation: It will enable inventors, businesses, and legal professionals to efficiently analyze patent documents and extract valuable insights; bridge the gap between patented inventions and commercial applications by identifying potential licensing opportunities and strategic partnerships Overall it will promote innovation by accelerating the pace of innovation by facilitating knowledge dissemination and collaboration within the patent ecosystem.



Project 20

Domain: Generative AI, Computer Vision, and Digital Heritage Preservation

Techniques: GANs, Image Super-Resolution, Neural Style Transfer, 3D Reconstruction

Title: AI-Powered Heritage Restoration: Reviving Historical Artifacts with Generative Models

Description:

This project focuses on using Generative AI and computer vision to restore damaged cultural heritage artifacts such as paintings, sculptures, and photographs. By applying advanced image processing techniques like GAN-based inpainting, super-resolution, and neural style transfer, the system can reconstruct missing details and enhance artifacts, preserving them for future generations in both physical and digital formats.

Overview:

Cultural heritage is at constant risk of damage due to aging, environmental factors, and improper handling. This project aims to build an AI-powered tool to digitally restore and enhance these artifacts, filling gaps caused by damage and improving resolution for better archival and display purposes. Using Generative Adversarial Networks (GANs) for texture synthesis and style transfer, the system mimics the original artist's style while maintaining authenticity.

Dataset:

1. **Metropolitan Museum of Art Open Access Images**
(<https://www.kaggle.com/datasets/metmuseum/the-metropolitan-museum-of-art-open-access/data>)
2. **Universe Roboflow dataset** (<https://universe.roboflow.com/art-dataset/wiki-art/dataset/2>)
3. Over 500,000 high-resolution images of art and artifacts from various historical periods. **WikiArt Dataset** (<https://www.wikiart.org/>)
4. Comprehensive dataset of artworks categorized by style, genre, and artist. **COCO Dataset for Image Completion** (<https://cocodataset.org/#overview>)
5. Annotated images for training inpainting and restoration models.
(<https://www.kaggle.com/datasets/defactodataset/defactoinpainting/data>)

Methodology:

- **Data Preprocessing:** Collect and clean high-resolution images of damaged and intact artifacts.



- **Image Inpainting with GANs:** Use models like DeepFill or StyleGAN to reconstruct missing parts of the artifact.
- **Super-Resolution:** Enhance the resolution of low-quality images using SRGANs or ESRGANs.
- **Style Transfer for Authenticity:** Use neural style transfer to apply the artifact's original texture and color patterns.
- **3D Reconstruction (Optional):** Convert 2D restored images into 3D models using depth estimation and mesh generation techniques.
- **Evaluation:** Use expert feedback and quantitative metrics like SSIM (Structural Similarity Index) for evaluation.

Project Deliverables:

1. A tool for digital restoration of cultural artifacts, including high-resolution images.
2. A dataset of restored artifacts for academic and cultural use.
3. Optional: 3D-rendered models of restored sculptures or monuments.

Challenges:

- **Authenticity vs. Creativity:** Ensuring that the restoration aligns with historical accuracy while utilizing AI-generated content.
- **Generalization:** Training models to work effectively across different styles, textures, and levels of degradation.

Significance:

1. Preserves cultural heritage for future generations by creating digital backups of damaged artifacts.
2. Assists historians, archivists, and researchers in studying the original art styles and details.
3. Democratizes access to historical artifacts by providing restored versions to museums and institutions worldwide.
4. Reduces the cost of manual restoration processes.
5. Promotes AI's role in creative industries and cultural conservation.

Reference:

1. Pathak, D., et al. (2016). "Context Encoders: Feature Learning by Inpainting." (<https://arxiv.org/abs/1604.07379>)
2. Ledig, C., et al. (2017). "Photo-Realistic Single Image Super-Resolution Using a Generative Adversarial Network (SRGAN)." (<https://arxiv.org/abs/1609.04802>)



3. Gatys, L. A., et al. (2016). "Image Style Transfer Using Convolutional Neural Networks." (https://www.cv-foundation.org/openaccess/content_cvpr_2016/papers/Gatys_Image_Style_Transfer_CVPR_2016_paper.pdf)
4. Metropolitan Museum of Art Open Access Initiative: <https://www.metmuseum.org/>
5. WikiArt Dataset: <https://www.wikiart.org/>
6. Introduction to image inpainting with a practical example from the e-commerce industry (<https://medium.com/data-science-at-microsoft/introduction-to-image-inpainting-with-a-practical-example-from-the-e-commerce-industry-f81ae6635d5e>)



Project 21

Domain: Healthcare and Bioinformatics

Techniques: Graph Neural Networks (GNN), Deep Learning, Data Visualization

Title: Drug Interaction Prediction using Graph Neural Networks

Description: Design a system that predicts potential interactions between drugs based on molecular data represented as graphs. The system can leverage GNNs to identify harmful interactions, providing a critical tool for drug safety evaluations.

Overview:

This project aims to design a drug interaction prediction system using Graph Neural Networks (GNNs) to evaluate molecular interactions. By representing drugs and their properties as graphs, the system can efficiently identify harmful interactions and potential side effects. The integration of deep learning enhances the predictive capability of GNNs, leveraging molecular graph structures for accurate modeling. This project provides valuable tools for drug safety evaluations, improving pharmaceutical development processes. Ultimately, it can minimize adverse drug reactions, saving lives and optimizing healthcare practices.

Dataset:

The Open Drug Knowledge Graph (ODKG) dataset

(<https://www.kaggle.com/datasets/mannbrinson/open-drug-knowledge-graph/data>)

contains detailed information about drugs, their properties, and potential interactions. It includes molecular structure data, drug-target relationships, and annotations, represented in a graph format. This dataset is well-suited for tasks involving graph-based predictions, such as identifying drug-drug interactions or predicting unknown relationships. Its structured format makes it compatible with GNN-based models, providing a strong foundation for healthcare and bioinformatics research.

Methodology:

- **Data Preprocessing:** Clean and preprocess the dataset to extract drug molecular graphs and features like atom types and bond structures.
- **Graph Representation:** Convert drug data into graph structures where nodes represent atoms and edges represent bonds.
- **Model Development:** Build a Graph Neural Network (e.g., Graph Convolutional Network or Graph Attention Network) to learn molecular interactions.
- **Feature Engineering:** Incorporate additional features such as drug-target relations or chemical properties into the graph representation.



- **Model Training and Validation:** Train the GNN on known drug interaction data and validate its performance using metrics like accuracy, precision, and recall.
- **Deployment:** Develop a user-friendly interface or API for drug interaction predictions, enabling its use in clinical or pharmaceutical settings.

Project Deliverables:

1. A trained GNN-based predictive model capable of identifying potential drug interactions with high accuracy.
2. A visualization tool or dashboard for displaying interaction predictions and their associated risks in an interpretable manner.

Challenges:

- **Data Sparsity:** Drug interaction datasets often have incomplete or imbalanced data, leading to challenges in model training.
- **Complexity of Molecular Structures:** Accurately encoding complex molecular graphs with all relevant features can be computationally expensive and time-consuming.

Significance:

This project can significantly enhance drug safety by identifying harmful drug interactions early in the development process. It provides a critical tool for pharmaceutical companies and healthcare professionals to mitigate risks associated with multi-drug therapies. By leveraging advanced GNNs, it ensures a more accurate understanding of molecular interactions compared to traditional methods. The system promotes faster, data-driven decision-making in drug development and approval processes. Ultimately, this project contributes to improved public health outcomes and reduced costs associated with adverse drug reactions.

Reference:

1. [Open Drug Knowledge Graph Dataset](#)
2. Kipf, T., & Welling, M. (2016). "Semi-Supervised Classification with Graph Convolutional Networks" (arXiv preprint).
3. Battaglia, P. W., et al. (2018). "Relational Inductive Biases, Deep Learning, and Graph Networks." Nature.
4. Zitnik, M., et al. (2018). "Modeling Polypharmacy Side Effects with Graph Convolutional Networks." Bioinformatics.
5. Wu, Z., et al. (2020). "A Comprehensive Survey on Graph Neural Networks." IEEE Transactions on Neural Networks and Learning Systems.



Project 22

Domain: Industrial Automation and Computer Vision

Techniques: Computer Vision, edge computing, real time processing

Title: VisionEdge: Real-Time Defect Detection and Classification in High-Speed Manufacturing Lines

Overview and Problem Statement:

Quality control in high-speed manufacturing lines often relies on manual inspection, which is prone to errors, slow, and resource-intensive. This project aims to automate defect detection and classification using computer vision and deep learning, enabling real-time processing, integration into industrial systems, and improved accuracy. The system will use high quality images/videos, identify anomalies such as scratches or misalignments, classify defects into categories, and provide actionable feedback.

Specific Challenges:

- Real-time image processing in high-speed production environments.
- Handling imbalanced datasets with rare defect samples.
- Adapting to environmental variations like lighting or camera angles.

Data Description:

- **MVTec Anomaly Detection Dataset:** Includes 15 industrial products, 6612 images, and 73 classes of defects.
 - Source: (<https://www.kaggle.com/datasets/ipythonx/mvtec-ad/data>)
- **DAGM 2007 Dataset:** Contains 18,200 images for defect detection on textured surfaces, including defect-free samples.
 - Source: (<https://www.kaggle.com/datasets/mhskjelvareid/dagm-2007-competition-dataset-optical-inspection/data>)

Methodology:

- Preprocess images (noise reduction, contrast enhancement), and apply data augmentation techniques such as flipping and noise injection.
- Train CNNs (e.g., ResNet, EfficientNet) for defect detection and YOLOv8 or Faster R-CNN for locating and classifying defects.
- Use TensorRT or ONNX for edge deployment and tune hyperparameters for maximum accuracy.
- Develop interfaces to integrate with manufacturing systems (MES, QMS) and deploy via APIs like Gradio or Streamlit with a real-time dashboard.



Technology Stack:

1. Python C++, Computer Vision Libraries: OpenCV, scikit-image
2. Deep Learning Frameworks: TensorFlow or PyTorch
3. Image Processing: Pillow, Scipy
4. Data Manipulation: NumPy, Pandas
5. **Optional:** Industrial Automation: OPC-UA libraries for integration with industrial systems

Modeling Techniques:

1. Convolutional Neural Networks (CNNs), YOLOv8 or Faster R-CNN for defect detection and classification
2. Anomaly detection techniques such as autoencoders or one-class SVMs
3. Traditional computer vision techniques like edge detection, blob analysis, and template matching
4. Ensemble methods combining multiple models for robust defect detection

Project Deliverables:

1. A trained and deployed computer vision model for defect detection and classification.
2. **Optional:** Real-time integration with a manufacturing line (or a simulation).
3. A user-friendly interface for visualizing defects and generating reports.
4. Comprehensive project documentation, including dataset details, codebase, and user manual.

Applications:

- Automated quality control in manufacturing, reduction of waste through precise defect detection, cost savings by replacing labor-intensive manual inspections.

References:

- MVTec Anomaly Detection Dataset (<https://www.kaggle.com/datasets/ipythonx/mvtec-ad/data>)
- DAGM 2007 Dataset (<https://www.kaggle.com/datasets/mhskjelvareid/dagm-2007-competition-dataset-optical-inspection/data>)
- “Deep Learning for Automated Visual Inspection in Manufacturing” (<https://www.mdpi.com/2571-5577/7/1/11>)



Project 23

Domain: Enterprise Compliance & Risk Management

Techniques: Prompt Engineering, RAG, Fine-tuning or Instruction Tuning of LLMs

Title: Automating Compliance Documentation with Generative AI

Overview and Problem Statement:

Organizations across regulated industries (e.g., finance, healthcare, telecom) are required to comply with complex and ever-evolving regulatory frameworks like GDPR, HIPAA, SOC 2, ISO 27001, PCI-DSS, etc. Manually creating, updating, and validating compliance documentation is time-consuming, costly, and error-prone.. The goal of the project is to build a Generative AI-based pipeline that can automatically generate compliance reports or documentation from structured logs, checklists, audit trails, and unstructured meeting notes, while ensuring adherence to regulatory formats and policies.

Dataset Description:

You may use the following datasets or generate similar (synthetic) data:

- **Data Sources**
 - Public regulatory frameworks:
 - General Data Protection Regulation: [EU GDPR Portal](#)
 - GDPR-articles-dataset: <https://www.kaggle.com/datasets/iuliabunescu23/gdpr-articles-dataset>
 - GDPR Violations: <https://www.kaggle.com/datasets/jessemostipak/gdpr-violations>
 - NIST SP 800-53 Rev. 5: a catalog of security and privacy controls for information systems and organizations. <https://csrc.nist.gov/publications/detail/sp/800-53/rev-5/final>
 - Enterprise templates or audit logs (synthetic or anonymized):
 - Incident logs
 - Risk assessment checklists
 - Internal control spreadsheets

Methodology:

1. Data Preparation:
 - Collect/simulate audit logs, checklist items, unstructured notes, preprocess into structured formats.



2. RAG Pipeline:

- Use FAISS/ChromaDB to store compliance documents, enable retrieval of relevant clauses or regulatory content.

3. Prompt Engineering & LLM Orchestration:

- Prompt LLMs (e.g., GPT-4, Llama-3) to fill templates, summarize audit logs, suggest clauses and evidence links

4. NER & Text Classification:

- Extract control-related entities (e.g., risk type, control number), categorize documentation sections.

5. Evaluation & Validation:

- Use SME validation (manual), Use BLEU/ROUGE for summaries, add a confidence score layer

Project Deliverables:

- A working GenAI pipeline (Jupyter/Streamlit app)
- API to input raw audit data and generate compliance documents
- Fine-tuned prompts/templates for GDPR/HIPAA
- Project report with system architecture, results, and limitations
- Documentation and demo video

Applications:

- Automated generation of audit-ready compliance reports, support for vendor risk assessments, tooling for startups to rapidly become audit-ready

Specific Challenges:

- Lack of labeled datasets (need to simulate data)
- Variability in compliance formats and templates across domains

References:

1. [EU GDPR Regulations](#)
2. [NIST SP 800-53 Rev. 5](#)
3. [Google Cloud Whitepaper on Compliance Automation](#)
4. [NLP-based Automated Compliance Checking of Data Processing Agreements against GDPR](#)



Project 24

Domain: Legal Systems

Techniques: Prompt Engineering, Zero-Shot/Few-Shot Learning, RAG

Title: Zero-Shot Legal Contract Clause Generator Using Domain-Aware LLMs

Overview and Problem Statement:

Legal contract drafting is a tedious, manual, and error-prone process that demands domain expertise and consistency in language. Organizations waste considerable time drafting or revising boilerplate clauses (e.g., confidentiality, indemnity, governing law) for NDAs, SLAs, or MoUs. So, the goal of the project is to design a zero-shot legal clause generator that can generate legally compliant and context-aware contract clauses from intent-based prompts, previous contracts, or user instructions—without requiring large-scale supervised training.

Dataset Description:

You may use the following datasets and/or simulated content:

- CUAD (Contract Understanding Atticus Dataset)
13,000+ expert-labeled clauses across 500+ contracts
Categories: Confidentiality, Governing Law, Termination, Indemnity, etc.
[CUAD GitHub Repo](#)
- Legal-Clause-Dataset
<https://www.kaggle.com/datasets/bahushruth/legalclausedataset>
- Simulated Contracts
Team can create synthetic NDAs, SLAs with marked clause types for prompt testing

Methodology:

1. **Data Preparation**
 - Parse contracts and segment into clause-level units, label common clause types and create a prompt-intent dictionary
2. **Prompt Design**
 - Design prompts like: "Generate a liability clause for a SaaS contract under California law.", "Provide a confidentiality clause for a joint R&D agreement."
3. **LLM Inference (Zero/Few-Shot)**
 - Use GPT-4 / Claude / DocLLM to generate or adapt clauses, optionally use LangChain + FAISS for retrieval-enhanced generation
4. **Evaluation**



- Use domain expert feedback (manual evaluation), automated metrics: BLEU/ROUGE for style similarity, LegalQA scoring (if available)

5. Optional Fine-Tuning (Advanced)

- Fine-tune LLaMA or open-source LLMs using CUAD clause types, build few-shot learning demos for complex clause types

Project Deliverables:

- Interactive clause generator (Streamlit or Jupyter app)
- Predefined prompts for key clause types (Confidentiality, Indemnity, etc.)

Applications:

- Legal contract automation tools for enterprises and law firms, Intelligent clause suggestion in document editors (like MS Word plugins), support for startups to create basic contracts without legal overhead

Specific Challenges:

- Legal hallucination: LLMs may generate plausible but incorrect legal language
- Jurisdiction-specific legal variations (e.g., U.S. vs EU laws)
- Evaluating clause legality and enforceability is non-trivial
- Sensitive nature of real contract data (anonymization may be required)
- Lack of zero-shot benchmarking datasets for legal clauses

References:

1. [DocLLM: Structured Document Understanding](#)
2. [LegalBench: A Benchmark for Legal Reasoning in LLMs](#)



Project 25

Domain: Generative AI R&D, Prompt Engineering Automation

Techniques: Prompt Templating, Embedding Similarity Search, RLHF, Few-shot and Zero-shot Prompt Refinement, Prompt Evaluation

Title: PromptPilot: An AI Copilot for Intelligent Prompt Design and Optimization in GenAI Workflows

Overview and Problem Statement:

The goal of this project is to design effective prompts for GenAI tasks. Currently, it is often trial-and-error based, highly subjective, and dependent on user experience. Inconsistent or poor prompts lead to suboptimal outputs, especially in enterprise use-cases like document generation, data analysis, or customer support automation. So, you need to build an AI-powered Prompt Engineering Copilot that learns from prior prompt-response pairs, suggests optimized prompts for similar tasks, and improves GenAI outputs in productivity tools.

Dataset Description:

To train and evaluate the AI-Driven Prompt Engineering model, you can use the following datasets.

1. Open Datasets for Prompt Learning - PromptSource (BigScience P3 Dataset):
Thousands of prompt templates for multiple NLP datasets.
(<https://huggingface.co/datasets/bigscience/P3>)
Download using:

```
from datasets import load_dataset  
dataset = load_dataset("bigscience/P3")
```
2. FLAN Collection (Instruction-Tuned Prompt Dataset): Used to fine-tune instruction-following models like FLAN-T5 (<https://huggingface.co/datasets/tatsu-lab/alpaca>)
(Unofficial FLAN-style) - Alternative FLAN-tuned dataset: (<https://github.com/google-research/FLAN>)
3. Prompt Collections from Awesome-LLM and Prompt Engineering Repos: Prompt engineering libraries with structured templates for coding, writing, summarization, etc. ([Awesome ChatGPT Prompts](#)), [Prompt Engineering Guide](#)
4. Feedback Data - Helpful Instructions Dataset (HH-RLHF)-



- GitHub Repository by Anthropic: Anthropic has made the HH-RLHF dataset available on GitHub. <https://github.com/anthropics/hh-rlhf>
- Oxen.ai Dataset Repository : Oxen.ai hosts the HH-RLHF dataset, providing an alternative download option. <https://www.oxen.ai/Anthropic/hh-rlhf/dir/main/>
- ModelScope Dataset Portal: ModelScope offers a mirror of the HH-RLHF dataset, accessible for download. <https://modelscope.cn/datasets/Anthropic/hh-rlhf>

Methodology:

1. **Prompt Database Construction:** Store prompts, responses, feedback, task type metadata; Use FAISS or ChromaDB for semantic similarity search
2. **Prompt Suggestion Engine:** Retrieve similar prompts based on task, embedding similarity; Rank prompts using prior feedback, usage frequency, or success score
3. **Copilot Interface (via LangChain):** Real-time recommendations for improving user prompts; Allow few-shot examples or template completions; Use LangChain memory for session context
4. **Feedback Loop Integration:** Let users rate prompt usefulness; retrain or re-rank suggestions; Store session history for personalized refinement
5. **Optional Add-ons:** Auto-labeling of task type; RLHF-style feedback weight adjustment; Prompt categorization/classification using BERT or LLMs

Project Deliverables:

- A web-based or Jupyter-based Prompt Copilot Tool
- Vector DB of prompts with FAISS/Chroma integration
- Interactive prompt suggestion module with history recall
- Feedback capture and re-ranking logic
- Evaluation report on prompt improvement effectiveness
- Documentation, screencast demo, and walkthrough

Applications:

- Productivity enhancement for GenAI users (enterprise, developers, analysts)
- Support teams designing chatbots, RAG systems, and customer interaction flows
- Internal AI task management or AI agent task decompositions

Specific Challenges:

- Defining a universal prompt quality metric is hard (subjective outcomes)



- Feedback collection and incorporating it into model suggestions
- Preventing prompt overfitting or loss of creativity due to templating
- Ensuring the tool works across different LLM backends

References:

1. [From Prompt Chaos to Clarity: Why I Built PromptPilot](#)
2. [PromptPilot](#)
3. [OpenAI Cookbook: Prompt Engineering Techniques](#)
4. [Awesome-LLM – Prompt Libraries](#)
5. [FAISS for Vector Search](#)



Project 26

Domain: Aviation Safety, Time-Series Analytics, Applied Machine Learning

Techniques: Supervised Machine Learning, Signal Processing, Eddy Dissipation Rate (EDR) Estimation

Title: Machine Learning Enabled Turbulence Prediction Using Flight Data for Safety Analysis

Overview and Problem Statement:

Air turbulence is a major factor in in-flight discomfort, structural fatigue, and safety-related incidents. With advancements in aircraft sensor systems and recording technologies, large volumes of flight operation data are now available. This project focuses on developing a predictive model that leverages such data to accurately forecast in-flight turbulence events. The goal of this project is to proactively alert pilots or systems to potential turbulence zones, thereby improving overall safety and reducing operational risks.

In the current flight scenario, the turbulence is an event which is not possible to predict sufficiently before it occurs. In stormy or heavy cloudy situations, it is difficult to know prior to the occurrence of turbulence during flight. So, in this project we will be using historical flight data and thereby we need to estimate turbulence severity using Eddy Dissipation Rate (EDR). The model must be generalizable across aircraft types and flight phases, using a minimal set of reliable parameters.

Dataset and Description:

1. Airplane accident-severity-prediction: <https://github.com/chawla201/Airplane-Accident-Severity-Prediction>
2. OpenSky Network – ADS-B Flight Trajectories:
ADS-B offers important trajectory data (altitude, speed, heading) which can align with turbulence events (<https://opensky-network.org/datasets/impala>)
3. EUREC 4A Catalog: <https://eurec4a.aeris-data.fr/>

Methodology:

- Preprocessing: Remove null/corrupted columns, apply correlation analysis
- Feature Engineering: Sliding window segmentation, flattening to time-series vectors
- Label Generation: Estimate EDR using vertical wind speed and true airspeed via WINDGRAD algorithm
- Modeling: Train multiple ML models (Random Forest, LSTM) and evaluate performance
- Validation: Cross-validation across different aircraft and flight phases



Project Deliverables:

- Cleaned and labeled turbulence dataset, Python-based modeling and evaluation pipeline
- Interactive visualization dashboard (optional) and Final report detailing model, metrics, and insights

Applications:

- Real-time turbulence warning systems in commercial aviation, Pilot advisory systems and safety dashboards
- Air traffic management decision support and Maintenance and fatigue analysis

Specific Challenges:

- Variability across aircraft types and atmospheric conditions, Imbalanced dataset with rare turbulence events
- Noise and missing data in flight recordings, Interpreting multi-sensor time-series data in real time

References:

1. EUREC4A turbulence dataset (<https://essd.copernicus.org/articles/13/3379/2021/>)
2. [Estimating Eddy Dissipation Rate with QAR Flight Big Data](#)
3. [Machine Learning Enabled Turbulence Prediction Using Flight Data](#)
4. [A Machine Learning-Based Model for Flight Turbulence Identification Using LiDAR Data](#)
5. [CAT-Predictor-ML: High-Altitude Clear Air Turbulence Forecasting](#)



Project 27

Domain: Environmental Sustainability, Computer Vision, Artificial Intelligence

Techniques: Image Classification, Object Detection (YOLOv5/YOLOv8/YOLOv11/Faster R-CNN), Convolutional Neural Networks (CNN), Transfer Learning, Data Augmentation, OpenCV.

Title: Smart Vision System for Plastic Waste Detection Using Deep Learning

Overview and Problem Statement:

Plastic waste has become one of the most pressing environmental threats. Manual sorting is labor-intensive and often ineffective. Optical methods powered by computer vision offer a scalable, real-time solution to detect and classify plastic waste in natural and urban environments. This project aims to build a deep learning-based model that can identify and categorize plastic waste from images captured via drones, CCTV cameras, or handheld devices. The goal of the project is to develop a robust computer vision model capable of detecting plastic waste in real-time images or video feeds to assist automated waste segregation and support environmental clean-up operations.

Dataset Description:

The project will utilize publicly available datasets such as:

1. Plastic-Detection-Model: https://github.com/antiplasti/Plastic-Detection-Model/blob/master/training_dataset.zip
2. Universe- Plastic Detection (v3 2023-11-15 5:53pm): <https://universe.roboflow.com/asia-pacific-university-msg4d/plastic-detection-o3dr4/dataset/3/download/yolov11>
3. Plastic Object Detection Dataset: This dataset contains images of various plastic objects commonly found in everyday life. Each image is annotated with bounding boxes around the plastic items, allowing for object detection tasks in computer vision applications. (<https://www.kaggle.com/datasets/dataclusterlabs/plastic-object-detection-dataset>)
4. TACO Dataset: <https://www.kaggle.com/datasets/kkhandekar/taco-dataset>
5. Custom image datasets gathered from Google Open Images and aerial surveillance datasets (if needed). Images will be labeled into categories like Plastic Bottles, Plastic Bags, Wrappers, etc.

Methodology:

- Data Preprocessing: Label cleanup, resizing, and augmentation (rotation, brightness, cropping).



- Model Selection: Use YOLOv5/YOLOv8/YOLOv11 or EfficientDet for object detection; compare performance with CNN-based classifiers.
- Training: Fine-tune on the plastic waste dataset using transfer learning on pre-trained COCO weights.
- Evaluation: Use metrics like mAP (mean Average Precision), precision, recall, and F1-score.
- Deployment: Build a lightweight interface using OpenCV or Streamlit to test on real-time video or camera feed.

Project Deliverables:

- Trained model capable of detecting plastic waste in diverse scenes.
- Visualization dashboard with bounding boxes for detected items.
- Evaluation report with accuracy metrics.
- Optional: Web or mobile-based demo prototype for live detection.

Applications:

- Automated plastic segregation in waste management plants.
- Real-time monitoring using drones in urban clean-up drives.
- Integration into smart city surveillance for pollution monitoring.
- Support for NGOs and civic bodies in environmental protection.

Specific Challenges:

- Varying lighting and weather conditions in outdoor images.
- Occlusion and overlapping of waste items.
- Differentiating between plastic and similar-looking non-plastic items.
- Ensuring lightweight and fast inference for real-time use.

References:

1. TACO Dataset: <https://www.kaggle.com/datasets/kkhandekar/taco-dataset>
2. YOLOv5 Documentation: <https://github.com/ultralytics/yolov5>
3. [Underwater Plastic Garbage Detection || YOLOv8](#)
4. [Deep learning-based waste detection in natural and urban environments](#)
5. [A Survey on Waste Detection and Classification Using Deep Learning](#)
6. OpenCV Documentation: <https://docs.opencv.org/>